Scalability & Future Plan: Architectural Evolution & Roadmap

Project: NeuroClass — Multimodal Attendance & Engagement Agent

Module: `8.Project Demonstration`

Phase: System Scalability, Enterprise Extensibility & Future Roadmap

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1. Architectural Scalability Strategy

NeuroClass was designed with a decoupled, modular foundation that allows seamless scaling from a single laptop setup in a 30-person seminar room to campus-wide multi-camera amphitheater deployments:

SCALABILITY & HORIZONTAL EXPANSION
PHASE 1: CURRENT STATE PHASE 2: NEAR-TERM EXPANSION PHASE 3: ENTERPRISE ECOSYSTEM
• Single-camera edge setup • Multi-camera spatial stitching • Full LTI 1.3 LMS integration
• Local SQLite ACID database • Edge NPU hardware acceleration • Campus-wide edge fleet cluster
• 30-50 participant tracking • 100-250 student lecture halls • Predictive curriculum analytics

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2. Key Evolutionary Roadmap Pillars

The ongoing architectural evolution of NeuroClass is organized around four strategic roadmap pillars targeting large-scale institutional adoption and low-power hardware acceleration.

Pillar 1: Multi-Camera Spatial Stitching for Tiered Amphitheaters

- **Challenge:** Large university lecture halls (200-500 seats) have tiered rows, blind spots, and steep viewing angles that exceed a single wide-angle lens.
- **Planned Solution:**
- Introduce multi-RTSP camera ingestion with homography-based coordinate alignment.
- Global re-identification (ReID) embedding vectors will track students across camera transitions as they move between room sections.

Pillar 2: Edge NPU & Hardware Acceleration (ONNX / OpenVINO)

- **Challenge:** Scaling to high-resolution 4K feeds and 100+ simultaneous face tracks increases CPU utilization.
- **Planned Solution:**
- Export MediaPipe and face detection models to ONNX and OpenVINO formats.
- Leverage on-chip Neural Processing Units (NPUs) such as Intel Core Ultra AI Boost, Apple Neural Engine, and Google Coral Edge TPUs to achieve 60 FPS at ***<10\text{W}*** power consumption.

Pillar 3: Learning Management System (LMS) Integration (LTI 1.3)

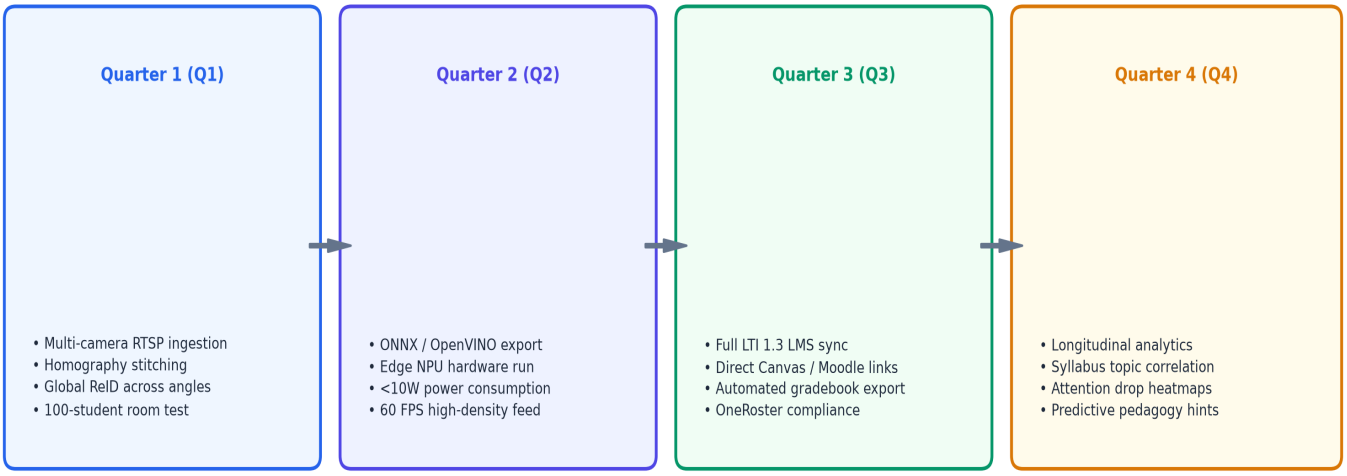
- **Challenge:** Universities require automated data synchronization with existing enterprise portals.
- **Planned Solution:**
- Implement Learning Tools Interoperability (LTI 1.3) and OneRoster standards.
- Verified attendance records and focus curves will automatically sync with **Canvas**, **Moodle**, **Blackboard**, and **Brightspace** gradebooks without manual CSV exports.

Pillar 4: Predictive Curriculum Analytics & Syllabus Optimization

- **Challenge:** Identifying curriculum topics that are systematically difficult across multiple cohorts.
- **Planned Solution:**
- Correlate time-stamped engagement drops against specific course syllabus modules.
- Provide semester-end pedagogical diagnostics highlighting topics that caused the sharpest attention drops, allowing professors to refine their teaching material iteratively.

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3. Technology Migration Roadmap (18-Month Plan)



4. Conclusion

The future of NeuroClass lies in deepening its capabilities as an **ethical, hardware-efficient edge platform** that empowers educators with actionable pedagogical intelligence while defending student privacy as an inviolable standard.